

Priors, Posteriors & the Beta–Binomial

INFO 521 · student notes · Module 4 · Lecture A · ~7 min read

Module 3 measured uncertainty by imagining repeated datasets, with the parameter a fixed unknown. Bayes flips what is random: keep the one dataset you have, and treat the **parameter itself** as uncertain — a distribution updated by evidence.

Bayes' rule

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta) p(\theta) \quad (\text{posterior} \propto \text{likelihood} \times \text{prior}).$$

The **prior** is belief before the data; the **likelihood** is the same Module 3 object; the **posterior** is belief after. Frequentist and Bayesian differ only in what is random — the estimate versus the belief about the parameter — and share the likelihood. The running question: what fraction θ of adults are hypertensive? Each screened patient is a Bernoulli draw, so after N patients with n_1 positives the likelihood is binomial in shape, $\theta^{n_1}(1 - \theta)^{n_0}$. The full-NHANES answer is $\theta \approx 0.388$.

The Beta prior and conjugacy

A prior for a proportion lives on $[0, 1]$; the **Beta** family fits, with parameters a, b read as **pseudo-counts** (“as if I had already seen $a - 1$ positives and $b - 1$ negatives”). Beta(1, 1) is flat — total ignorance.

The key mechanism is **conjugacy**: multiplying the Beta prior by the binomial likelihood adds the exponents, so

$$\text{Beta}(a, b) \xrightarrow{n_1 \text{ ones, } n_0 \text{ zeros}} \text{Beta}(a + n_1, b + n_0).$$

The posterior stays in the family, and updating collapses to arithmetic: count successes, add to a ; count failures, add to b . Watching real NHANES draws, the posterior narrows around 0.388 as patients accumulate; and three very different priors converge to nearly the same posterior after 200 patients — priors matter most when data are scarce, but enough evidence washes them out. A bad prior delays convergence, it does not destroy it.

What the posterior buys

The posterior is a complete answer. A **point estimate** is its mean $\frac{a+n_1}{a+b+N}$ (or mode); an **error bar** is a 95% **credible interval** read off the density (there is a 95% probability θ lies in it — what people wish a confidence interval meant). The mean is a weighted **blend** of prior mean and sample frequency, with the data's weight approaching 1 as N grows. Prediction integrates over parameter uncertainty (the **posterior predictive**) rather than plugging in a point estimate; with a flat prior this is Laplace's rule of succession, $\frac{n_1+1}{N+2}$, never exactly 0 or 1.

The Beta–Binomial explorer lets you set the prior (a, b) , add “positive” and “negative” patients one at a time, and watch the posterior mean and variance update while the posterior curve shifts.

What to remember

- Posterior $\propto$ likelihood $\times$ prior; Bayes makes the *parameter* the random object.
- Beta–Binomial conjugacy makes the update pure addition on the pseudo-counts.
- One posterior yields point estimates, credible intervals, and predictions.
- Priors dominate when data are scarce and fade as evidence accumulates.

Through-lines

Everything here is one-dimensional by design. Lecture B runs the same conjugate trick on the *regression weights* — a Gaussian prior meets a Gaussian likelihood — and that closed form is where Part 1 ends.